

LOGOVOY, Kazakhstan

WMO: 383430

Lat: 42.950N

Lon: 72.750E

Elev: 2241

StdP: 13.54

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	0.0	-9.8	3.5	-5.8	-4.5	4.7	0.2	12.0	27.0	10.0	27.1	0.7	0	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.6	96.0	65.0	93.3	64.0	90.7	63.2	66.9	89.0	65.6	88.0	64.6	86.6	4.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	82.4	74.5	57.7	77.3	73.8	56.2	73.0	73.1	32.7	88.7	31.7	88.7	30.9	87.0	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.9	10.1	8.6	-9.9	100.9	8.3	2.4	-15.9	102.6	-20.8	104.1	-25.5	105.4	-31.5	107.2		
			-10.4	70.0	8.1	2.8	-16.2	72.0	-21.0	73.6	-25.5	75.1	-31.4	77.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.3	24.6	28.3	43.2	53.8	62.9	72.0	76.0	73.6	63.9	51.4	38.7	26.2
	DBStd	20.09	11.21	11.75	9.66	8.08	6.40	5.65	4.86	5.55	7.45	8.66	10.41	11.38
	HDD50	2878	787	607	246	52	1	0	0	0	2	85	359	739
	HDD65	5957	1252	1027	675	342	117	11	1	3	109	427	789	1204
	CDD50	3368	0	0	36	166	400	659	806	733	420	129	19	0
	CDD65	973	0	0	0	5	51	220	342	271	77	7	0	0
	CDH74	13166	0	0	8	142	702	2797	4642	3618	1113	136	8	0
	CDH80	5883	0	0	1	28	168	1185	2345	1727	395	33	1	0
	WSAvg	2.9	2.2	2.6	3.4	3.5	3.5	3.5	3.4	3.1	2.8	2.5	2.3	2.1
	PrecAvg	15.0	0.7	1.0	1.4	2.2	2.2	1.5	1.1	0.7	0.6	1.3	1.3	1.0
Precipitation	PrecMax	22.1	1.9	2.1	2.6	4.9	4.7	3.6	3.1	2.0	1.8	3.7	3.2	2.0
	PrecMin	8.2	0.2	0.3	0.4	0.4	0.8	0.3	0.2	0.1	0.1	0.0	0.0	0.1
	PrecStd	3.5	0.4	0.4	0.5	1.1	0.9	0.8	0.7	0.5	0.5	0.9	0.8	0.5
	PrecStd	3.5	0.4	0.4	0.5	1.1	0.9	0.8	0.7	0.5	0.5	0.9	0.8	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	60.1	73.8	83.6	88.2	96.9	100.0	98.4	92.3	85.0	73.1	54.1
		MCWB	46.1	50.2	59.3	63.3	63.8	65.2	66.3	65.3	62.7	60.3	55.8	45.4
	2%	DB	46.6	52.4	67.8	78.0	84.4	93.0	96.6	95.4	88.7	77.3	64.8	48.5
		MCWB	40.3	44.8	54.3	60.3	62.6	64.6	65.6	64.3	61.2	57.0	52.2	41.7
	5%	DB	42.1	47.5	63.5	73.6	81.2	89.9	93.7	92.5	84.8	72.6	58.5	44.0
		MCWB	37.3	41.6	52.4	58.8	61.6	63.7	64.3	63.4	60.1	54.8	48.2	38.5
	10%	DB	38.4	43.0	58.9	69.2	77.9	86.9	91.1	89.3	80.8	67.4	53.6	40.2
		MCWB	34.8	38.4	50.0	56.7	60.8	62.9	63.5	62.5	58.4	52.3	45.9	35.8
	0.4%	WB	46.4	50.2	59.3	64.6	66.8	68.1	69.2	67.5	64.4	61.2	56.9	45.8
		MCDB	54.3	59.8	73.0	80.4	82.6	88.9	90.0	92.5	86.3	83.0	71.5	53.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.9	45.1	55.2	61.4	64.2	66.3	67.0	65.6	62.4	57.8	53.0	42.0
		MCDB	45.7	51.6	66.5	75.5	80.5	86.8	90.2	90.4	85.5	75.5	63.3	48.3
	5%	WB	37.7	41.7	52.7	59.3	62.7	65.1	65.7	64.5	60.7	55.4	49.2	38.6
		MCDB	41.6	46.7	62.4	72.7	78.6	85.4	89.4	89.0	82.9	70.5	57.6	43.5
	10%	WB	34.8	38.5	50.2	57.2	61.2	64.0	64.6	63.3	59.0	53.1	45.8	36.0
		MCDB	38.4	42.9	58.4	68.4	76.3	83.8	87.8	86.7	79.0	66.3	52.9	40.0
Mean Daily Temperature Range	5% DB	MDBR	16.3	16.3	19.4	20.8	23.3	25.0	26.6	27.6	27.3	23.6	18.5	16.3
		MCDBR	21.5	23.1	27.5	29.7	28.7	29.9	30.7	30.8	31.9	31.8	27.8	22.4
		MCWBR	16.6	17.0	17.0	16.2	13.0	10.9	10.8	10.8	13.0	16.4	16.9	17.1
	5% WB	MCDBR	20.3	22.1	26.5	28.0	25.9	26.8	27.9	28.8	30.7	28.8	25.9	21.6
		MCWBR	15.9	16.7	16.9	15.8	13.0	11.2	10.9	11.1	13.1	15.4	16.0	16.6
Clear-Sky Solar Irradiance	taub		0.351	0.394	0.462	0.487	0.464	0.461	0.452	0.417	0.397	0.428	0.400	0.344
	taud		2.098	2.014	1.907	1.899	2.031	2.072	2.106	2.192	2.191	2.016	2.028	2.112
	Ebn at Noon		235	245	242	248	259	259	260	266	261	229	212	225
	Edh at Noon		33	43	54	59	53	51	49	43	40	43	36	30
All-Sky Solar Radiation	RadAvg		630	921	1246	1601	2005	2201	2217	2068	1636	1021	636	514
	RadStd		49	78	91	146	141	120	128	105	84	117	72	43

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.35	-3.28	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.94	N/A	N/A	N/A	N/A	

Nomenclature: See separate page